

Jin Young Ahn

andy0808opp@snu.ac.kr | +82 10-3304-4621 | Seoul, Korea | <https://ahnjinyoung.github.io/>

EDUCATION

Seoul National University, Seoul, South Korea

(expected graduation date : 2027.02)

Major Department of Computer Science and Engineering

Relevant Coursework:

Data Structures, Computer Architecture, Operating systems, System Programming, Algorithms, NLP, Artificial intelligence, Intro to Deep learning, Linear Algebra, Abstract Algebra, Engineering Mathematics, Statistics, Hardware system Design(TPU)

GPA : 3.44 / 4.3

PERSONAL INFORMATION

Military status : Completed military service as Sergeant, Republic of Korea Army

RESEARCH INTEREST

Robot foundation models, Diffusion models, Long-horizontal generation, Math for ML

EXPERIENCE

Research Intern, Prof. Namjoon Kim's Lab, Seoul National University

2025.12 – Present

- Built **CoMoE**, a Mixture-of-Experts LLM trained from scratch (2B total / 500M active parameters), outperforming published MoE architectures at comparable scale (UMoE, MoA) under a matched training budget
- Reproduced the π_0 vision-language-action model on an xArm7 manipulator: built the teleoperation setup and collected the demonstration dataset used for fine-tuning

PUBLICATIONS

GyuYeop Do*, Jiho Ryou*, Jinwoo Jeon, **Jin Young Ahn**, Sunwoong Kim, Hyuk-Jae Lee†, Hyungon Ryu†, Nam-Joon Kim†. "CoMoE: Collaborative Expert Aggregation for Sparse Mixture-of-Experts." *Under review, EMNLP 2026*.

* Equal contribution † Corresponding author

PROJECT

On-device Vision Model Optimization on Edge TPU (with LG)

- Optimizing on-device vision models for deployment on Edge TPU hardware (industry collaboration with LG)
- Model quantization, compilation, and latency/accuracy trade-off analysis for real-time inference
- HW Platform : Raspberry Pi 5, Google coral Edge TPU

FastMRI challenge (In progress)

- Reconstruct undersampled MRI images into fully-sampled MRI images
- Achieved > 0.93 SSIM accuracy with size of < 30M parameters in Knee MRI image reconstruction

SKILLS

- Programming language : C, Python
- Experienced skills :
 - Pytorch, RISC-V assembly, Java, Rust, Scala, Docker, Kubernetes, Zero-DP
 - Running pi0 in xArm7 (single arm, DoF 7) , single arm teleoperation
- *Language Skills*: English(Fluent) , Korean (Native)